The Complete Treatise on the Macroeconomic Framework of Planet Earth

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Abstract

This treatise develops the foundations of a macroeconomic framework for planet Earth. Unlike conventional macroeconomics, the framework begins from heterogeneous geology, endogenous extraction technology, and a structurally closed monetary–geopolitical manifold.

Physical resources, extraction technology, monetary institutions, state capacity, and geopolitical power are modelled as components of a single interacting dynamical system.

The resulting framework provides a unified foundation for resource economics, macroeconomics, finance, political economy, international relations, warfare economics, data science, and artificial intelligence.

The treatise ends with "The End"

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Chapter 1

Foundations

1.1 Introduction

Most macroeconomic models begin with aggregate output.

The present framework begins instead with Earth itself.

Economic activity ultimately depends upon:

1. physical resources,
2. extraction technology,
3. labor,
4. capital,
5. institutions,
6. monetary systems,
7. geopolitical structures.

Consequently, macroeconomics may be viewed as an emergent layer built upon geological and technological foundations.

The objective of this treatise is to construct a mathematically integrated framework linking:

$$Geology \to Technology \to \begin{matrix} Resource \\ Extraction \end{matrix} \to Macroeconomy \to \begin{matrix} State \\ Capacity \end{matrix} \to Geopolitics \to \begin{matrix} Monetary \\ Structure \end{matrix}.$$

1.2 Planet Earth as an Economic Object

Let Earth be represented as a sphere of radius

$$R > 0.$$

The total volume is

$$V = \frac{4}{3}\pi R^3.$$

Let

$$\rho_P(r,t)$$

denote the physical density field.
The total physical mass satisfies

$$M = \int_0^R 4\pi \rho_P(r,t) r^2 dr.$$

Remark 1. *Physical mass is a geological quantity rather than an economic quantity. Economic significance emerges only through recoverability and valuation.*

1.3 The Structurally Closed Monetary–Geopolitical Manifold

Definition 1. *A structurally closed monetary–geopolitical manifold is a state space*

$$\mathcal{M}$$

whose coordinates describe the monetary, political, institutional, technological, strategic, and military conditions of a civilization.

A representative point is

$$x = (m, g, s, \tau, \ell, \eta, T_E, \ldots).$$

Components may include:

- monetary conditions,
- geopolitical conditions,
- strategic conditions,
- institutional conditions,

- logistics,
- military capability,
- extraction technology.

1.4 Extraction Technology

Definition 2. *Extraction technology stock is*

$$T_E(t) \geq 0.$$

Examples include:

- drilling technologies,
- mining technologies,
- geological imaging,
- autonomous systems,
- robotics,
- artificial intelligence,
- advanced materials.

Remark 2. *Technology is treated as a state variable rather than merely a productivity coefficient.*

1.5 Recoverability

Not all resources can be extracted.

Define recoverability

$$\Gamma(x,t) \in [0,1].$$

The economically recoverable density field is

$$\boxed{\rho(x,t) = \Gamma(x,t)\rho_P(x,t).}$$

Proposition 1. *If*

$$\frac{\partial \Gamma}{\partial T_E} > 0,$$

then technological progress increases recoverable resources.

Proof. Since

$$\rho = \Gamma \rho_P,$$

the result follows immediately. □

1.6 Geological Embedding

Define an embedding

$$\Phi : [0, R] \to \mathcal{M}.$$

Thus

$$r \mapsto x(r).$$

Every geological location is associated with a monetary–geopolitical state.

1.7 Price and Cost Fields

1.7.1 Price Field

Define

$$p(x, t).$$

Prices depend upon:

- resource abundance,
- monetary conditions,
- geopolitical conditions,
- technological conditions.

1.7.2 Cost Field

Define

$$c(x, t).$$

Costs depend upon:

- wages,

- interest rates,
- logistics,
- regulations,
- technology,
- security conditions.

1.8 Margin Field

Definition 3. *The margin field is*

$$m(x,t) = p(x,t) - c(x,t).$$

1.9 Technology-Dependent Accessibility

Let

$$r_{\max}(x)$$

denote the radius of the deepest inaccessible region.
The extractable shell is

$$r_{\max}(x) \leq r \leq R.$$

Assume

$$\frac{\partial r_{\max}}{\partial T_E} < 0.$$

Improved technology permits deeper extraction.

1.10 The Fundamental Profit Functional

Theorem 1 (Planetary Profit Functional). *Planetary profit satisfies*

$$\boxed{\Pi = \int_{r_{\max}(x)}^{R} 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t) r^2 dr.}$$

Proof. Substituting

$$\rho = \Gamma \rho_P$$

and

$$m = p - c$$

into the standard extraction profit equation yields the result. □

1.11 Technology Elasticities

Define

$$\varepsilon_p = \frac{\partial \ln p}{\partial \ln T_E},$$

$$\varepsilon_c = \frac{\partial \ln c}{\partial \ln T_E},$$

$$\varepsilon_\Gamma = \frac{\partial \ln \Gamma}{\partial \ln T_E}.$$

These quantities measure technological sensitivity.

Remark 3. *Technological progress may increase or decrease profits depending upon the relative magnitudes of these elasticities.*

1.12 Resource Wealth

Define resource wealth

$$W(t) = \Pi(t).$$

This quantity forms the bridge between geology and macroeconomics.

1.13 Technology Accumulation

Extraction technology evolves according to

$$\boxed{\dot{T}_E = \eta_E R_E - \delta_E T_E.}$$

where

- R_E = extraction R&D,
- η_E = innovation efficiency,
- δ_E = technological depreciation.

1.14 Resource Discoveries

Technology generates discoveries.

Define

$$D(T_E, x).$$

The resource stock evolves according to

$$\boxed{\dot{S} = D(T_E, x) - q.}$$

where

$$q$$

is the extraction rate.

1.15 Physics

Extraction requires energy.

Define

$$\varepsilon(x, t).$$

Total extraction energy is

$$E_{\text{req}} = \int_{r_{\max}(x)}^{R} 4\pi\varepsilon(x, t)\rho(x, t)r^2 dr.$$

Remark 4. *Thermodynamics imposes lower bounds on extraction costs.*

1.16 Chemistry

Define concentration field

$$\chi(x, t).$$

Recoverable resource quantity becomes

$$Q = \int_{r_{\max}(x)}^{R} 4\pi\chi(x,t)\rho(x,t)r^2 dr.$$

Chemical technologies influence recovery rates and refinement efficiencies.

1.17 Biology

Resource extraction affects ecological systems.

Let

$$B$$

denote biodiversity.

Assume

$$B = B(E),$$

where

$$E$$

is environmental disturbance.

Typically

$$\frac{dB}{dE} < 0.$$

1.18 Psychology and Expectations

Economic agents form expectations regarding:

- future prices,
- future costs,
- future technologies,
- geopolitical conditions.

Expected future profit is

$$E_t[\Pi_{t+1}].$$

Expectations influence investment and exploration.

1.19 Political Science

Resource wealth affects state capacity.

Let

$$S_C$$

denote state capacity.

Assume

$$S_C = S_0 + \alpha W.$$

Resource extraction therefore influences governance capabilities.

1.20 International Relations

Resource wealth affects international influence.

Let

$$G_P$$

denote geopolitical power.

Assume

$$G_P = G_0 + \beta W.$$

Resource-rich systems may acquire greater strategic influence.

1.21 Warfare Economics

Military capability satisfies

$$M_S = M_0 + \gamma W.$$

Military capability and resource wealth therefore co-evolve.

1.22 The Closed Feedback Loop

The framework generates the following cycle:

$$Technology \to Recoverability \to Profit \to \begin{matrix} State \\ Capacity \end{matrix} \to Geopolitics \to Prices \ \& \ Costs \to Profit.$$

1.23 Integrated State Representation

The complete state vector is

$$z = (x, T_E, S, W, S_C, G_P, M_S).$$

Macroeconomics emerges from the evolution of this state.

1.24 System Architecture

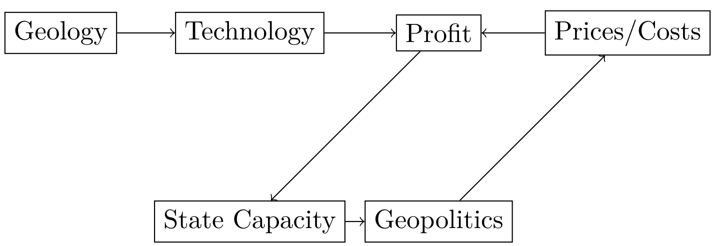

Figure 1.1: System architecture.

1.25 Summary Table

Symbol	Interpretation
$\mathcal{M}$	Monetary–geopolitical manifold
x	Manifold state
T_E	Extraction technology
Γ	Recoverability
ρ_P	Physical density
ρ	Recoverable density
p	Price field
c	Cost field
m	Margin field
Π	Profit
W	Resource wealth
S	Resource stock
S_C	State capacity
G_P	Geopolitical power
M_S	Military capability

Table 1.1: Foundational variables.

1.26 Conclusion

Part I establishes the foundational architecture of the macroeconomic framework of planet Earth.

The central result is the Planetary Profit Functional

$$\Pi = \int_{r_{\max}(x)}^{R} 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t)r^2 dr,$$

which links geology, technology, monetary structures, and geopolitical dynamics.

Subsequent parts will derive aggregate production, capital accumulation, monetary dynamics, equilibrium conditions, welfare economics, finance, statistical estimation, machine learning, and optimal control.

Chapter 2

Macroeconomics

2.1 Aggregate Macroeconomics

2.1.1 Introduction

Part I established the geological, technological, monetary, and geopolitical foundations of the framework.

The next objective is to derive aggregate macroeconomic relationships consistent with:

1. heterogeneous geology,
2. endogenous extraction technology,
3. recoverability,
4. manifold-valued prices,
5. manifold-valued costs,
6. dynamic resource stocks.

Unlike conventional macroeconomic models, aggregate output is not primitive.

Instead, it emerges from the interaction of resource extraction, technology, capital, labor, and institutions.

2.2 The Resource Base of Production

Let

$$\Pi(t)$$

denote planetary resource profit:

$$\Pi = \int_{r_{\max}(x)}^{R} 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t)r^2 dr.$$

This quantity measures the economically recoverable surplus generated by Earth.

Definition 4. *Planetary resource wealth is*

$$W(t) = \Pi(t).$$

2.3 Aggregate Production

Let

- $K(t)$ denote physical capital,
- $L(t)$ denote labor,
- $T_E(t)$ denote extraction technology,
- $\Pi(t)$ denote resource profit.

Definition 5 (Planetary Production Function)**.** *Aggregate output satisfies*

$$\boxed{Y = A(x,t)K^{\alpha}L^{\beta}\Pi^{\gamma}T_E^{\nu}.}$$

Here:

- $A(x,t)$ denotes manifold productivity,
- $\alpha, \beta, \gamma, \nu > 0$.

Remark 5. *Unlike Solow-style frameworks, resource profitability enters directly as a productive input.*

2.4 Output Elasticities

Proposition 2. *Output elasticities satisfy*

$$\frac{\partial \ln Y}{\partial \ln K} = \alpha,$$

$$\frac{\partial \ln Y}{\partial \ln L} = \beta,$$

$$\frac{\partial \ln Y}{\partial \ln \Pi} = \gamma,$$

$$\frac{\partial \ln Y}{\partial \ln T_E} = \nu.$$

Proof. Take logarithms:

$$\ln Y = \ln A + \alpha \ln K + \beta \ln L + \gamma \ln \Pi + \nu \ln T_E.$$

Differentiate with respect to each variable. □

2.5 Returns to Scale

Definition 6. *The production system exhibits constant returns to scale whenever*

$$\alpha + \beta + \gamma + \nu = 1.$$

Increasing returns occur when

$$\alpha + \beta + \gamma + \nu > 1.$$

Decreasing returns occur when

$$\alpha + \beta + \gamma + \nu < 1.$$

2.6 Capital Accumulation

Capital evolves according to

$$\boxed{\dot{K} = I - \delta K.}$$

Here

- I = investment,
- δ = depreciation.

2.6.1 Investment Function

Suppose

$$I = sY + \eta \Pi.$$

Then

$$\boxed{\dot{K} = sY + \eta \Pi - \delta K.}$$

Resource profits therefore finance additional capital formation.

2.7 Labor Dynamics

Let labor evolve according to

$$\boxed{\dot{L} = nL.}$$

Hence

$$L(t) = L_0 e^{nt}.$$

Population growth therefore influences productive capacity.

2.8 Technology Dynamics

From Part I,

$$\boxed{\dot{T}_E = \eta_E R_E - \delta_E T_E.}$$

Technology is an endogenous state variable.

2.9 Knowledge Production

Suppose research expenditure satisfies

$$R_E = \theta_Y Y + \theta_\Pi \Pi.$$

Then

$$\dot{T}_E = \eta_E(\theta_Y Y + \theta_\Pi \Pi) - \delta_E T_E.$$

Resource wealth therefore finances technological progress.

2.10 Resource Stock Dynamics

Resource stocks satisfy

$$\boxed{\dot{S} = D(T_E, x) - q.}$$

where

- D = discoveries,
- q = extraction.

Remark 6. *Unlike Hotelling-style depletion models, resource stocks may increase through discovery.*

2.11 The Government Sector

Government revenue consists of:

1. production taxes,
2. resource taxes.

Define

$$\tau_Y$$

and

$$\tau_\Pi.$$

Revenue becomes

$$\boxed{G_R = \tau_Y Y + \tau_\Pi \Pi.}$$

2.12 Government Expenditure

Let

$$G = G_C + G_I + G_M,$$

where:

- G_C = civilian spending,
- G_I = infrastructure,
- G_M = military spending.

2.13 State Capacity

State capacity satisfies

$$\boxed{S_C = S_0 + \alpha_G G_R.}$$

Thus

$$\frac{\partial S_C}{\partial \Pi} = \alpha_G \tau_\Pi.$$

Resource wealth strengthens administrative capability.

2.14 The Monetary Sector

Let

$$M(t)$$

denote money supply.
Assume

$$\boxed{\dot{M} = \mu_M M + \phi \Pi.}$$

The parameter

$$\phi$$

captures the monetary impact of resource wealth.

2.15 Inflation

Let

$$P(t)$$

denote the price level.
Inflation is

$$\pi = \frac{\dot{P}}{P}.$$

Assume

$$\boxed{\pi = \lambda_1 \frac{M}{Y} + \lambda_2 \frac{G}{Y} - \lambda_3 \frac{\Pi}{Y}.}$$

Resource abundance may mitigate inflationary pressures.

2.16 Interest Rates

Let

$$i(t)$$

denote the interest rate.
Suppose

$$i = i_0 + \psi_1 \pi - \psi_2 \frac{\Pi}{Y}.$$

Resource-rich economies may enjoy lower financing costs.

2.17 The Financial Sector

2.17.1 Resource Asset Valuation

The value of resource assets is

$$\boxed{F = \sum_{t=0}^{\infty} \frac{E[\Pi_t]}{(1+i)^t}.}$$

Remark 7. *Resource wealth becomes capitalized into financial markets.*

2.18 The External Sector

Define:

- X_R = resource exports,
- X_O = non-resource exports,
- IM = imports.

Net exports:

$$\boxed{NX = X_R + X_O - IM.}$$

2.19 National Income Identity

Aggregate output satisfies

$$\boxed{Y = C + I + G + NX.}$$

2.20 Resource Dependence

Define

$$\theta_R = \frac{\Pi}{Y}.$$

Interpretation:

- $\theta_R \approx 0$ implies diversification,
- $\theta_R \approx 1$ implies extreme dependence.

2.21 Resource Stabilization Funds

Let

$$F_S$$

denote a stabilization fund.
Suppose

$$\dot{F}_S = \omega\Pi - \zeta F_S.$$

Resource rents may therefore be transformed into long-term financial assets.

2.22 Intergenerational Wealth

Define comprehensive wealth:

$$\Omega = K + F_S + V_S.$$

Here

$$V_S$$

denotes remaining resource wealth.
Sustainability requires

$$\dot{\Omega} \geq 0.$$

2.23 Macroeconomic Equilibrium

Definition 7. *A macroeconomic equilibrium is a state*

$$(K^*, L^*, T_E^*, S^*, M^*)$$

such that

$$\dot{K} = \dot{L} = \dot{T}_E = \dot{S} = \dot{M} = 0.$$

2.24 Steady-State Technology

Theorem 2. *If*

$$R_E = \bar{R}_E,$$

then

$$T_E^* = \frac{\eta_E}{\delta_E} \bar{R}_E.$$

Proof. At equilibrium,

$$\dot{T}_E = 0.$$

Hence

$$\eta_E \bar{R}_E = \delta_E T_E^*.$$

Solving yields the result. □

2.25 Dynamic System

Collecting the equations yields

$$\boxed{\dot{K} = sY + \eta\Pi - \delta K}$$

$$\boxed{\dot{L} = nL}$$

$$\boxed{\dot{T}_E = \eta_E R_E - \delta_E T_E}$$

$$\boxed{\dot{S} = D(T_E, x) - q}$$

$$\boxed{\dot{M} = \mu_M M + \phi \Pi}$$

These equations define the macroeconomic core.

2.26 The Planetary Growth Mechanism

Growth arises through five channels:

1. capital accumulation,
2. labor accumulation,
3. technological progress,
4. resource discoveries,
5. geopolitical productivity.

Thus

$$\text{Growth} = f(K, L, T_E, S, x).$$

2.27 Summary Table

Variable	Interpretation
Y	Aggregate output
K	Capital stock
L	Labor
T_E	Extraction technology
S	Resource stock
M	Money supply
G_R	Government revenue
S_C	State capacity
F	Resource asset value
F_S	Stabilization fund
θ_R	Resource dependence ratio

Table 2.1: Macroeconomic variables.

2.28 Conclusion

Part II derives the aggregate macroeconomic structure of planet Earth.

The key innovation is that resource profitability, extraction technology, and monetary–geopolitical conditions become endogenous components of aggregate production and growth.

The resulting framework extends traditional macroeconomics by embedding production, finance, government, monetary systems, and external balances within a geological–technological foundation.

Part III will develop political economy, state formation, geopolitical competition, international relations, military capability, and warfare economics.

Chapter 3

Political Economy and Geopolitics

3.1 Political Economy, State Formation, Geopolitics, and Warfare Economics

3.1.1 Introduction

Part II established the aggregate macroeconomic structure of planet Earth.

The present part develops the political and strategic layers of the framework.

The central thesis is that:

$$\text{Resources} \to \text{Revenue} \to \text{State Capacity} \to \text{Military Capability} \to \text{Geopolitical Influence}.$$

Political institutions, military organizations, and international relations are therefore viewed as emergent structures built upon geological, technological, and macroeconomic foundations.

3.2 Resource Wealth and Political Economy

Recall that planetary resource profit is

$$\Pi = \int_{r_{\max}(x)}^{R} 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t)r^2 dr.$$

Resource wealth is

$$W = \Pi.$$

This quantity forms the fiscal foundation of political systems.

3.3 The State

Definition 8. *A state is an organization possessing:*

1. *taxation authority,*
2. *legal authority,*
3. *coercive authority,*
4. *administrative authority.*

Let

$$S_C$$

denote state capacity.

3.4 State Capacity Function

Assume

$$\boxed{S_C = S_0 + \alpha_Y Y + \alpha_\Pi \Pi.}$$

State capacity therefore depends upon both productive output and resource wealth.

Proposition 3. *If*

$$\alpha_\Pi > 0,$$

resource wealth increases state capacity.

Proof. Differentiate:

$$\frac{\partial S_C}{\partial \Pi} = \alpha_\Pi.$$

□

3.5 Institutional Quality

Let

$$I_Q$$

denote institutional quality.

Examples include:

- legal systems,
- property rights,
- bureaucratic competence,
- regulatory quality.

Assume

$$I_Q = I_Q(S_C).$$

Typically

$$\frac{dI_Q}{dS_C} > 0.$$

3.6 Tax Capacity

Government revenue was

$$G_R = \tau_Y Y + \tau_\Pi \Pi.$$

Suppose effective taxation satisfies

$$\tau_Y = \tau_Y(I_Q),$$

$$\tau_\Pi = \tau_\Pi(I_Q).$$

Higher institutional quality improves fiscal extraction.

3.7 Political Stability

Let

$$P_S$$

denote political stability.
Assume

$$P_S = f(Y, S_C, I_Q).$$

Then

$$\frac{\partial P_S}{\partial Y} > 0,$$

$$\frac{\partial P_S}{\partial S_C} > 0.$$

Economic development and administrative competence increase stability.

3.8 Resource Dependence and Political Risk

Recall

$$\theta_R = \frac{\Pi}{Y}.$$

Assume political risk satisfies

$$R_P = R_0 + \kappa\theta_R.$$

Large resource dependence may increase political risk.

3.9 State Formation Dynamics

Let

$$N(t)$$

denote state organizational complexity.
Assume

$$\dot{N} = \eta_N S_C - \delta_N N.$$

Administrative structures therefore evolve through time.

3.10 Military Capability

Definition 9. *Military capability is*

$$M_S.$$

Assume

$$\boxed{M_S = M_0 + \beta_G G_M + \beta_T T_M.}$$

where

- G_M = military expenditure,
- T_M = military technology.

3.11 Military Technology

Military technology evolves according to

$$\boxed{\dot{T}_M = \eta_M R_M + \xi T_E - \delta_M T_M.}$$

The parameter

$$\xi$$

captures civilian-to-military technological spillovers.
Examples include:

- artificial intelligence,
- robotics,
- satellites,
- advanced materials,
- autonomous systems.

3.12 Defense Allocation

Let

$$\delta_D \in [0, 1]$$

denote the defense share of government expenditure.
Then

$$\boxed{G_M = \delta_D G.}$$

Resource wealth therefore indirectly finances military capability.

3.13 Strategic Reserves

Let

$$R_S$$

denote strategic reserves.
Examples include:

- petroleum reserves,
- food reserves,
- rare-earth reserves,
- monetary reserves.

Assume

$$\dot{R}_S = \omega_S \Pi - \delta_S R_S.$$

3.14 Geopolitical Power

Define geopolitical power:

$$G_P.$$

Assume

$$\boxed{G_P = a_0 + a_1 S_C + a_2 M_S + a_3 \Pi.}$$

Remark 8. *Geopolitical influence depends simultaneously upon institutions, military capability, and resource wealth.*

3.15 International Relations

Suppose

$$n$$

states exist.
Let

$$G_i$$

denote geopolitical power of state i.
Interactions occur through:

- trade,
- diplomacy,
- alliances,
- sanctions,
- conflict.

3.16 Geopolitical Networks

Define adjacency matrix

$$A = (a_{ij}).$$

where

$$a_{ij} = \begin{cases} 1, & \text{interaction exists,} \\ 0, & \text{otherwise.} \end{cases}$$

The resulting graph describes geopolitical structure.

3.17 Alliance Formation

Suppose alliance value is

$$V_A.$$

Assume

$$V_A = \sum_{i=1}^{n} \omega_i G_i.$$

Alliances aggregate geopolitical power.

3.18 Trade Networks

Let

$$T_{ij}$$

denote trade flow from state i to state j.
Total trade volume:

$$T = \sum_i \sum_j T_{ij}.$$

Trade influences both economic output and geopolitical influence.

3.19 Sanctions

Let

$$\sigma_{ij}$$

denote sanction intensity.
Effective trade becomes

$$\widetilde{T}_{ij} = (1 - \sigma_{ij})T_{ij}.$$

Sanctions therefore alter network structure.

3.20 Conflict Economics

Suppose conflict yields benefit

$$B_C.$$

Conflict cost is

$$C_C.$$

Net conflict value:

$$V_C = B_C - C_C.$$

Conflict becomes attractive whenever

$$V_C > 0.$$

3.21 A Resource Conflict Model

Suppose two states compete over a resource region producing profit

$$\Pi_R.$$

Expected conflict value:

$$V_C = \lambda \Pi_R - C_C.$$

where

$$0 \leq \lambda \leq 1.$$

Conflict incentives increase with resource profitability.

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

Table 3.1: Payoff matrix.

3.22 Game-Theoretic Representation

Consider two states.

Actions:

- Cooperate (C),
- Compete (D).

Payoff matrix:

Resource competition may therefore be analyzed through game theory.

3.23 Warfare Economics

Military expenditure contributes to aggregate demand:

$$Y = C + I + G + NX.$$

Since

$$G = G_C + G_I + G_M,$$

military expenditure affects output directly.

3.24 Military Keynesian Effects

Assume

$$\frac{\partial Y}{\partial G_M} > 0.$$

Short-run output may therefore increase through defense expenditure.

3.25 Military Opportunity Costs

However,

$$G_M$$

reduces resources available for:

- infrastructure,
- education,
- research,
- healthcare.

Thus military spending possesses opportunity costs.

3.26 Resource Security

Define resource security index

$$\Psi.$$

Assume

$$\Psi = \Psi(R_S, M_S, S_C).$$

Then

$$\frac{\partial \Psi}{\partial R_S} > 0,$$

$$\frac{\partial \Psi}{\partial M_S} > 0.$$

Strategic reserves and military capability improve security.

3.27 Monetary Sovereignty

Let

$$M$$

denote money supply.
Monetary sovereignty depends upon:

- state capacity,
- geopolitical influence,
- resource wealth.

Define

$$\Sigma = \Sigma(S_C, G_P, \Pi).$$

3.28 The Geopolitical State Equation

The geopolitical subsystem evolves according to

$$\boxed{\dot{G}_P = F_G(S_C, M_S, \Pi, T, M).}$$

Geopolitical power is therefore endogenous.

3.29 The Political Economy State Equation

Political institutions evolve according to

$$\boxed{\dot{I}_Q = F_I(S_C, Y, \Pi).}$$

Institutions are dynamic rather than fixed.

3.30 The Strategic State Vector

The political-economic state becomes

$$z_P = (S_C, I_Q, G_P, M_S, R_S, \Sigma).$$

This vector summarizes the strategic condition of a civilization.

3.31 Integrated Political Economy System

The complete subsystem is

$$\boxed{S_C = S_0 + \alpha_Y Y + \alpha_\Pi \Pi}$$

$$\boxed{M_S = M_0 + \beta_G G_M + \beta_T T_M}$$

$$\boxed{G_P = a_0 + a_1 S_C + a_2 M_S + a_3 \Pi}$$

$$\boxed{\dot{T}_M = \eta_M R_M + \xi T_E - \delta_M T_M}$$

These equations define the political and strategic architecture of the framework.

3.32 Summary Table

Variable	Interpretation
S_C	State capacity
I_Q	Institutional quality
P_S	Political stability
M_S	Military capability
T_M	Military technology
G_P	Geopolitical power
R_S	Strategic reserves
Σ	Monetary sovereignty
R_P	Political risk
Ψ	Resource security

Table 3.2: Political economy and geopolitical variables.

3.33 Conclusion

Part III develops the political economy, geopolitical, and warfare-economic layers of the Macroeconomic Framework of planet Earth.

The principal result is that resource wealth influences state formation, institutional quality, military capability, strategic reserves, geopolitical influence, and monetary sovereignty.

The resulting system transforms geology and extraction technology into political and strategic power through a sequence of endogenous feedback mechanisms.

Part IV will develop welfare economics, environmental systems, physics, chemistry, biology, psychology, psychiatry, and the broader human consequences of planetary resource extraction.

Chapter 4

Multidisciplinary Framework

4.1 Welfare Economics, Environmental Systems, Physics, Chemistry, Biology, Psychology, and Psychiatry

4.1.1 Introduction

Parts I–III established the geological, technological, macroeconomic, political, and geopolitical architecture of planet Earth.

The next step is to analyze the broader consequences of resource extraction and economic development.

Economic activity does not occur in isolation.

Production, extraction, technological progress, military activity, and state formation all interact with:

1. physical systems,
2. chemical systems,
3. biological systems,
4. environmental systems,
5. psychological systems,
6. social systems.

Consequently, welfare analysis must be embedded within a multidisciplinary framework.

4.2 Social Welfare

Definition 10. *Social welfare is*

$$W = W(C, E, S_C, G_P, B, H),$$

where

- C *= consumption,*
- E *= environmental damage,*
- S_C *= state capacity,*
- G_P *= geopolitical security,*
- B *= biodiversity,*
- H *= human well-being.*

4.3 A Welfare Functional

A simple specification is

$$\boxed{W = U(C) - D(E) + \omega_S S_C + \omega_G G_P + \omega_B B + \omega_H H.}$$

The coefficients

$$\omega_S, \omega_G, \omega_B, \omega_H$$

represent social weights.

4.4 Intertemporal Welfare

Future welfare is discounted.

Define

$$0 < \beta < 1.$$

The social planner solves

$$\boxed{\max \sum_{t=0}^{\infty} \beta^t W_t.}$$

4.5 Comprehensive Wealth

Let

$$\Omega = K + F_S + V_S + N_C.$$

Components:

- K = physical capital,
- F_S = stabilization funds,
- V_S = resource wealth,
- N_C = natural capital.

Definition 11. *A system is sustainable whenever*

$$\dot{\Omega} \geq 0.$$

4.6 Environmental Systems

Resource extraction generates environmental disturbances.

Let

$$E(t)$$

denote cumulative environmental impact.

Assume

$$\boxed{\dot{E} = \phi_q q - \phi_T T_E.}$$

Interpretation:

- extraction increases damage,
- cleaner technologies reduce damage.

4.7 Climate Effects

Let

$$C_L$$

denote a climate-stress index.
Assume

$$\dot{C}_L = \alpha_E E - \delta_C C_L.$$

Environmental damage contributes to climate stress.

4.8 Environmental Carrying Capacity

Let

$$K_C$$

denote carrying capacity.
Assume

$$K_C = K_C(E).$$

Typically

$$\frac{dK_C}{dE} < 0.$$

Environmental degradation reduces carrying capacity.

4.9 Physics

4.9.1 Energy Requirements

Extraction requires energy.
Define

$$\varepsilon(x,t)$$

as energy per unit recoverable mass.
Total extraction energy is

$$\boxed{E_{\text{req}} = \int_{r_{\max}(x)}^{R} 4\pi\varepsilon(x,t)\rho(x,t)r^2 dr.}$$

4.10 Thermodynamic Constraints

Let

$$0 < \eta < 1$$

denote energetic efficiency.
Then

$$E_{\text{useful}} = \eta E_{\text{input}}.$$

Thus

$$E_{\text{input}} \geq E_{\text{req}}.$$

Theorem 3 (Thermodynamic Feasibility)**.** *Resource extraction is feasible only if*

$$E_{\text{input}} \geq E_{\text{req}}.$$

Proof. Immediate from the Second Law of Thermodynamics and the definition of efficiency. □

4.11 Geophysical Constraints

Let

$$T_G(r)$$

denote the geothermal temperature profile.
Let

$$P_G(r)$$

denote the pressure profile.
Extraction costs generally satisfy

$$\frac{\partial c}{\partial T_G} > 0,$$

$$\frac{\partial c}{\partial P_G} > 0.$$

Deep extraction therefore becomes progressively more difficult.

4.12 Chemistry

4.12.1 Resource Concentrations

Let

$$\chi_i(x,t)$$

denote concentration of resource i.
Examples include:

- iron,
- copper,
- lithium,
- uranium,
- rare earth elements.

Recoverable quantity:

$$Q_i = \int_{r_{\max}(x)}^{R} 4\pi\chi_i(x,t)\rho(x,t)r^2dr.$$

4.13 Recovery Efficiency

Let

$$0 \leq \gamma_i \leq 1.$$

Recovered quantity becomes

$$Q_i^{\text{rec}} = \gamma_i Q_i.$$

Technological progress may increase

$$\gamma_i.$$

4.14 Reaction Networks

Many extraction processes involve chemical transformations.

Let

$$\mathbf{c}(t)$$

denote concentration vector.

Reaction dynamics satisfy

$$\dot{\mathbf{c}} = F_C(\mathbf{c}).$$

This representation accommodates metallurgical and refining processes.

4.15 Biology

4.15.1 Biodiversity

Let

$$B(t)$$

denote biodiversity.

Assume

$$\boxed{\dot{B} = r_B B \left(1 - \frac{B}{K_B}\right) - \theta_E E.}$$

Environmental damage reduces biodiversity.

4.16 Agricultural Capacity

Let

$$A_G$$

denote agricultural productivity.

Assume

$$A_G = A_G(B, E).$$

Typically

$$\frac{\partial A_G}{\partial B} > 0,$$

$$\frac{\partial A_G}{\partial E} < 0.$$

4.17 Public Health

Let

$$H_P$$

denote public health.
Assume

$$H_P = H_P(Y, E, S_C).$$

Higher incomes and stronger institutions improve health outcomes.
Environmental degradation may worsen them.

4.18 Psychology

4.18.1 Expectations

Economic agents form expectations:

$$E_t[\Pi_{t+1}],$$

$$E_t[Y_{t+1}],$$

$$E_t[p_{t+1}].$$

These expectations influence:

- investment,
- consumption,
- exploration,
- innovation.

4.19 Behavioral Biases

Define behavioral distortion

$$B_D.$$

Examples:

- overconfidence,
- anchoring,
- recency bias,
- herd behavior.

Perceived profits become

$$\widetilde{\Pi} = \Pi + B_D.$$

4.20 Social Cohesion

Let

$$S_H$$

denote social cohesion.
Assume

$$S_H = S_H(Y, H_P, P_S).$$

Economic prosperity and political stability strengthen cohesion.

4.21 Psychiatry

4.21.1 Stress Dynamics

Let

$$\Sigma_P$$

denote social stress.
Assume

$$\boxed{\dot{\Sigma}_P = \alpha_U U + \alpha_\pi \pi + \alpha_C C_L - \delta_\Sigma \Sigma_P.}$$

Stress rises with:

- unemployment,
- inflation,
- climate stress.

4.22 Mental Health

Let

$$M_H$$

denote population mental health.
Assume

$$M_H = M_H(H_P, \Sigma_P, S_H).$$

Typically

$$\frac{\partial M_H}{\partial \Sigma_P} < 0.$$

4.23 Human Development

Define a development index

$$D_H.$$

Assume

$$D_H = D_H(Y, H_P, M_H, E).$$

Human development therefore depends upon economic, environmental, and psychological variables.

4.24 Ethics and Philosophy

The framework permits multiple normative objectives.
Examples include:

1. consumption maximization,
2. welfare maximization,
3. sustainability,

4. intergenerational equity,
5. capability expansion.

Different philosophical perspectives correspond to different welfare functionals.

4.25 The Social Planner Problem

The planner chooses:

$$q, \quad R_E, \quad G, \quad G_M, \quad F_S.$$

The objective is

$$\boxed{\max \sum_{t=0}^{\infty} \beta^t W_t.}$$

subject to:

- geological constraints,
- technological constraints,
- fiscal constraints,
- environmental constraints,
- geopolitical constraints.

4.26 Integrated Human-Earth System

The state vector becomes

$$z_H = (K, L, T_E, S, M, S_C, G_P, E, B, H_P, M_H).$$

The economy is therefore embedded within a larger human-Earth system.

4.27 Summary Table

Variable	Interpretation
W	Social welfare
E	Environmental damage
C_L	Climate stress
B	Biodiversity
A_G	Agricultural productivity
H_P	Public health
S_H	Social cohesion
Σ_P	Social stress
M_H	Mental health
D_H	Human development
Ω	Comprehensive wealth

Table 4.1: Human and environmental variables.

4.28 Conclusion

Part IV embeds the macroeconomic framework within physical, chemical, biological, environmental, psychological, psychiatric, and philosophical systems.

The central conclusion is that economic activity is not an isolated process. Resource extraction, technological progress, and geopolitical competition influence ecological systems, human well-being, public health, social cohesion, and long-run sustainability.

The resulting framework is therefore a coupled human-Earth system in which geology, technology, economics, politics, ecology, and psychology co-evolve through time.

Part V will develop statistics, econometrics, Bayesian learning, data science, machine learning, reinforcement learning, world models, digital twins, computational algorithms, and planetary-scale decision systems.

Chapter 5

Empirical Framework

5.1 Statistics, Econometrics, Data Science, Artificial Intelligence, and Planetary Decision Systems

5.1.1 Introduction

Parts I–IV established the geological, technological, macroeconomic, political, geopolitical, environmental, biological, and social foundations of the macroeconomic framework of planet Earth.

The present part develops the informational and computational layer.

The central premise is that no planner directly observes the complete state of planet Earth.

Instead, decision-making occurs under uncertainty.

Consequently:

1. statistics quantifies uncertainty,
2. econometrics estimates parameters,
3. machine learning identifies patterns,
4. artificial intelligence constructs policies,
5. world models generate forecasts,
6. digital twins support planetary-scale decisions.

5.2 The Statistical State Space

Let

$$z_t = (K_t, L_t, T_{E,t}, S_t, M_t, S_{C,t}, G_{P,t}, E_t, B_t, H_{P,t}, M_{H,t})$$

denote the latent planetary state.
The true state is generally unobservable.
Observed data satisfy

$$y_t = h(z_t) + \varepsilon_t.$$

Definition 12. *The mapping*

$$h$$

is the observation operator.

5.3 Parameter Estimation

Let

$$\theta$$

denote the parameter vector.
Examples include:

- production elasticities,
- technological coefficients,
- resource discovery parameters,
- geopolitical coefficients,
- environmental parameters.

5.3.1 Likelihood Estimation

Given data

$$D,$$

the likelihood is

$$L(\theta|D).$$

Parameter estimates satisfy

$$\boxed{\hat{\theta} = \arg\max_{\theta} L(\theta|D).}$$

5.4 Econometric Identification

Suppose

$$Y_t = f(X_t, \theta) + u_t.$$

Identification requires that distinct parameter vectors generate distinct observable implications.

Definition 13. *A parameter vector is identified if*

$$f(X, \theta_1) = f(X, \theta_2)$$

implies

$$\theta_1 = \theta_2.$$

5.5 Bayesian Learning

Uncertainty regarding parameters is represented by a prior

$$p(\theta).$$

After observing data,

$$D,$$

posterior beliefs become

$$\boxed{p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.}$$

Theorem 4 (Bayesian Updating)**.** *Posterior beliefs equal likelihood multiplied by prior and normalized.*

Proof. Direct application of Bayes' theorem. □

5.6 Geological Learning

Let

$$\theta_G$$

denote geological parameters.
Examples:

- density fields,
- concentration fields,
- recoverability fields,
- discovery probabilities.

Exploration data update beliefs:

$$p(\theta_G|D_G).$$

5.7 Entropy and Information

Let

$$G$$

denote geological uncertainty.
Entropy is

$$\boxed{H(G) = -\sum_i p_i \ln p_i.}$$

Exploration reduces uncertainty.

Proposition 4. *Information acquisition reduces entropy:*

$$H(G|I) \le H(G).$$

5.8 Monte Carlo Methods

Let

$$\omega_1, \ldots, \omega_N$$

denote simulated scenarios.
Expected profit is approximated by

$$\boxed{\widehat{\Pi} = \frac{1}{N}\sum_{i=1}^{N} \Pi(\omega_i).}$$

Theorem 5 (Monte Carlo Consistency). *As*

$$N \to \infty,$$

$$\widehat{\Pi} \to E[\Pi].$$

Proof. Law of Large Numbers. □

5.9 Time-Series Analysis

Planetary variables evolve through time.

Examples:

$$Y_t, \quad \Pi_t, \quad M_t, \quad G_{P,t}.$$

A generic autoregressive representation is

$$X_t = a_0 + a_1 X_{t-1} + \varepsilon_t.$$

5.10 State-Space Representation

The planetary system can be represented by

$$z_{t+1} = F(z_t, a_t, \varepsilon_t),$$

$$y_t = h(z_t) + u_t.$$

This representation underlies filtering and forecasting.

5.11 Data Science

The complete dataset may include:

- geological surveys,
- satellite imagery,
- commodity prices,
- financial data,

- military indicators,
- climate data,
- health data.

Define

$$X \in \mathbb{R}^{n \times p}.$$

The objective is to extract actionable information from high-dimensional data.

5.12 Supervised Learning

Suppose

$$X$$

denotes features.
Suppose

$$Y$$

denotes outcomes.
Learning seeks

$$Y = f(X) + \varepsilon.$$

Examples include:

- resource forecasting,
- price forecasting,
- conflict forecasting,
- growth forecasting.

5.13 Neural Networks

A neural network approximates

$$\hat{Y} = f_\theta(X).$$

Training solves

$$\boxed{\min_\theta \sum_i (Y_i - \hat{Y}_i)^2.}$$

5.14 Geological Neural Networks

Input features may include:

- seismic data,
- gravity data,
- magnetic data,
- drilling data.

Output variables may include:

- density estimates,
- concentration estimates,
- recoverability estimates.

5.15 Economic Forecasting Networks

Inputs:

- output,
- inflation,
- money supply,
- resource profits.

Outputs:

- growth forecasts,
- inflation forecasts,
- recession probabilities.

5.16 Geopolitical Prediction

Let

$$P_C$$

denote conflict probability.
A predictive model satisfies

$$P_C = f(G_P, M_S, \Pi, \Sigma).$$

Machine learning can estimate this function.

5.17 Reinforcement Learning

5.17.1 State

Define

$$s_t = z_t.$$

5.17.2 Actions

Actions include:

- extraction,
- investment,
- taxation,
- reserve accumulation,
- diplomacy,
- military allocation.

5.17.3 Reward

Define

$$R_t = W_t,$$

where

$$W_t$$

is welfare.

5.17.4 Objective

The planner solves

$$\boxed{\max_{\pi} E\left[\sum_{t=0}^{\infty} \beta^t R_t\right].}$$

5.18 Bellman Equation

Define value function

$$V(s).$$

Then

$$\boxed{V(s) = \max_a E\left[R + \beta V(s')\right].}$$

5.19 Q-Learning

Define

$$Q(s, a).$$

Optimal values satisfy

$$Q^*(s, a) = E\left[R + \beta \max_{a'} Q^*(s', a')\right].$$

5.20 Multi-Agent Reinforcement Learning

Suppose

$$n$$

states exist.
Each solves

$$\max E\left[\sum_t \beta^t R_t^i\right].$$

The resulting system captures:

- cooperation,

- competition,
- alliances,
- strategic conflict.

5.21 World Models

Let

$$z_t$$

denote latent planetary state.
World dynamics satisfy

$$\boxed{z_{t+1} = F(z_t, a_t, \varepsilon_t).}$$

The model predicts future states under alternative actions.

5.22 Digital Twin of Earth

Definition 14. *A digital twin is a computational replica of the planetary system.*

Components include:

1. geological model,
2. macroeconomic model,
3. monetary model,
4. geopolitical model,
5. environmental model,
6. AI planning model.

5.23 Planetary Optimization

The planner chooses:

$$q_t, \quad R_{E,t}, \quad G_t, \quad G_{M,t}, \quad F_{S,t}.$$

Subject to:

- resource constraints,
- technological constraints,
- fiscal constraints,
- environmental constraints,
- geopolitical constraints.

5.24 The Planetary Control Problem

The objective is

$$\boxed{\max E\left[\sum_{t=0}^{\infty} \beta^t W_t\right].}$$

This is the central optimization problem of the framework.

5.25 Pseudo-Code

5.25.1 Planetary Resource Planner

```
Initialize state z

Observe data

Estimate parameters

Update beliefs

Forecast future states

Evaluate candidate actions

Select welfare-maximizing action

Implement action

Observe outcomes

Repeat
```

5.26 Artificial General Planning Architecture

The architecture consists of:

1. observation layer,
2. estimation layer,
3. forecasting layer,
4. optimization layer,
5. execution layer.

This structure integrates statistics, machine learning, and decision theory.

5.27 Integrated Computational State

The complete computational state is

$$z_C = (z, \theta, D, I, V, Q).$$

where:

- z = planetary state,
- θ = parameters,
- D = data,
- I = information,
- V = value function,
- Q = action-value function.

5.28 The Planetary Information System

Combining Parts I–V yields:

$$\text{Geology} \to \text{Technology} \to \text{Economy} \to \text{Politics} \to \text{Environment} \to \text{Information} \to \text{Decision}.$$

Information therefore becomes an endogenous productive resource.

5.29 Summary Table

Variable	Interpretation
θ	Parameter vector
$H(G)$	Geological entropy
V	Value function
Q	Action-value function
z_t	Planetary state
D	Dataset
I	Information set
P_C	Conflict probability
R_t	Reward
π	Policy

Table 5.1: Computational and informational variables.

5.30 Conclusion

Part V develops the informational architecture of the macroeconomic framework of planet Earth.

The framework integrates:

- statistics,
- econometrics,
- Bayesian learning,
- information theory,
- data science,
- machine learning,
- reinforcement learning,
- world models,
- digital twins,
- planetary-scale optimization.

The resulting system transforms Earth into a stochastic, observable, learnable, and controllable dynamical system.

Part VI will present the integrated system of equations, equilibrium concepts, master tables, glossary, bibliography, and final synthesis of the complete framework.

Chapter 6

Single Mathematical Architecture

6.1 Integrated System, Equilibrium Theory, Master Tables, Glossary, and Conclusion

6.1.1 Introduction

Parts I–V developed the geological, technological, macroeconomic, political, geopolitical, environmental, biological, psychological, statistical, computational, and artificial-intelligence layers of the macroeconomic framework of planet Earth.

The purpose of this final part is to unify all previous components into a single mathematical architecture.

The resulting framework may be interpreted as a coupled:

1. geological system,
2. technological system,
3. macroeconomic system,
4. political system,
5. geopolitical system,
6. environmental system,
7. informational system,
8. decision system.

6.2 The Fundamental Geological Equation

The physical resource base is

$$M = \int_0^R 4\pi \rho_P(r,t) r^2 dr.$$

Recoverable density satisfies

$$\rho(x,t) = \Gamma(x,t)\rho_P(x,t).$$

Technology enters through the manifold state

$$x = (m, g, s, \tau, \ell, \eta, T_E, \ldots).$$

6.3 The Fundamental Profit Functional

The central profit equation of the framework is

$$\boxed{\Pi = \int_{r_{\max}(x)}^R 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t) r^2 dr.}$$

where

$$m(x,t) = p(x,t) - c(x,t).$$

Remark 9. *Since*

$$T_E \in x,$$

technology influences:

- *prices,*
- *costs,*
- *recoverability,*
- *accessibility.*

6.4 Technology Dynamics

Extraction technology evolves according to

$$\boxed{\dot{T}_E = \eta_E R_E - \delta_E T_E.}$$

Military technology evolves according to

$$\boxed{\dot{T}_M = \eta_M R_M + \xi T_E - \delta_M T_M.}$$

6.5 Resource Dynamics

Resource stocks evolve according to

$$\boxed{\dot{S} = D(T_E, x) - q.}$$

where

- $D =$ discoveries,
- $q =$ extraction.

6.6 Aggregate Production

Aggregate output is

$$\boxed{Y = A(x,t) K^{\alpha} L^{\beta} \Pi^{\gamma} T_E^{\nu}.}$$

Output depends simultaneously upon:

- capital,
- labor,
- technology,
- resource profitability,
- manifold productivity.

6.7 Capital Accumulation

Capital evolves according to

$$\boxed{\dot{K} = sY + \eta\Pi - \delta K.}$$

6.8 Labor Dynamics

Labor evolves according to

$$\boxed{\dot{L} = nL.}$$

6.9 Government Revenue

Government revenue is

$$\boxed{G_R = \tau_Y Y + \tau_\Pi \Pi.}$$

6.10 State Capacity

State capacity satisfies

$$\boxed{S_C = S_0 + \alpha_Y Y + \alpha_\Pi \Pi.}$$

6.11 Military Capability

Military capability satisfies

$$\boxed{M_S = M_0 + \beta_G G_M + \beta_T T_M.}$$

6.12 Geopolitical Power

Geopolitical influence is

$$\boxed{G_P = a_0 + a_1 S_C + a_2 M_S + a_3 \Pi.}$$

6.13 Money Supply

Monetary dynamics satisfy

$$\boxed{\dot{M} = \mu_M M + \phi \Pi.}$$

6.14 Inflation

Inflation satisfies

$$\boxed{\pi = \lambda_1 \frac{M}{Y} + \lambda_2 \frac{G}{Y} - \lambda_3 \frac{\Pi}{Y}.}$$

6.15 Environmental Dynamics

Environmental damage evolves according to

$$\boxed{\dot{E} = \phi_q q - \phi_T T_E.}$$

6.16 Biodiversity Dynamics

Biodiversity evolves according to

$$\boxed{\dot{B} = r_B B \left(1 - \frac{B}{K_B}\right) - \theta_E E.}$$

6.17 Social Stress

Social stress evolves according to

$$\boxed{\dot{\Sigma}_P = \alpha_U U + \alpha_\pi \pi + \alpha_C C_L - \delta_\Sigma \Sigma_P.}$$

6.18 The Welfare Functional

The welfare function is

$$\boxed{W = U(C) - D(E) + \omega_S S_C + \omega_G G_P + \omega_B B + \omega_H H.}$$

6.19 The Planner's Problem

The social planner chooses

$$q, \quad R_E, \quad G, \quad G_M, \quad F_S.$$

The objective is

$$\boxed{\max E\left[\sum_{t=0}^{\infty} \beta^t W_t\right].}$$

6.20 The Planetary State Vector

Define

$$z = (K, L, T_E, T_M, S, M, S_C, G_P, E, B, H_P, M_H).$$

This vector summarizes the planetary state.

6.21 Planetary Dynamics

The complete system may be written compactly as

$$\boxed{\dot{z} = F(z, a, \varepsilon).}$$

where

- a = actions,
- ε = stochastic shocks.

6.22 Deterministic Equilibrium

Definition 15. *A deterministic equilibrium is a state*

$$z^*$$

such that

$$\dot{z} = 0.$$

6.23 Existence of Equilibrium

Theorem 6 (Equilibrium Existence)**.** *Suppose:*

1. *the state space is compact,*
2. *policy sets are non-empty,*

3. transition functions are continuous.

Then at least one equilibrium exists.

Proof. By continuity and compactness, the induced state transition mapping satisfies the conditions of Brouwer's Fixed Point Theorem.

Therefore a fixed point exists. □

6.24 Stochastic Equilibrium

Definition 16. *A stochastic equilibrium is a probability distribution*

$$\mu^*$$

satisfying

$$\mu^* = P\mu^*,$$

where

$$P$$

is the transition operator.

6.25 Information and Learning

Beliefs evolve according to

$$\boxed{p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.}$$

The planetary system continuously learns from data.

6.26 World Models

The latent state evolves according to

$$\boxed{z_{t+1} = F(z_t, a_t, \varepsilon_t).}$$

World models approximate this mapping.

6.27 Digital Twin of Earth

A digital twin consists of:

1. geological module,
2. macroeconomic module,
3. monetary module,
4. geopolitical module,
5. environmental module,
6. AI planning module.

The digital twin generates policy forecasts.

6.28 Planetary Reinforcement Learning

The planner's value function is

$$\boxed{V(s) = \max_a E\left[R + \beta V(s')\right].}$$

The reward is social welfare.

6.29 Integrated System Architecture

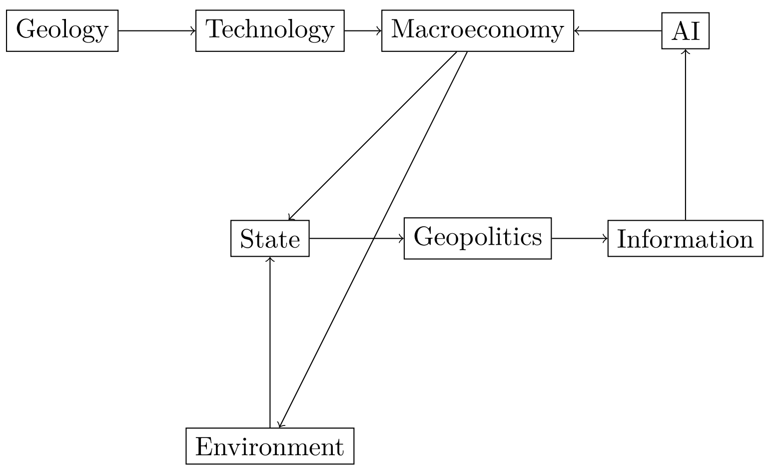

Figure 6.1: Integrated system architecture.

6.30 Master Variable Table

Variable	Interpretation
K	Capital stock
L	Labor
T_E	Extraction technology
T_M	Military technology
S	Resource stock
Π	Resource profit
Y	Output
M	Money supply
S_C	State capacity
M_S	Military capability
G_P	Geopolitical influence
E	Environmental damage
B	Biodiversity
H_P	Public health
M_H	Mental health
W	Social welfare

Table 6.1: Master state variables.

6.31 Master Equation System

The complete framework is summarized by

$$\boxed{\Pi = \int_{r_{\max}(x)}^{R} 4\pi m(x,t)\Gamma(x,t)\rho_P(x,t)r^2 dr}$$

$$\boxed{Y = A(x,t)K^\alpha L^\beta \Pi^\gamma T_E^\nu}$$

$$\boxed{\dot{T}_E = \eta_E R_E - \delta_E T_E}$$

$$\boxed{\dot{K} = sY + \eta\Pi - \delta K}$$

$$\boxed{\dot{S} = D(T_E, x) - q}$$

$$\boxed{\dot{z} = F(z, a, \varepsilon)}$$

These equations constitute the mathematical core of the framework.

6.32 Final Conclusion

The macroeconomic framework of planet Earth begins with heterogeneous geology and culminates in a planetary-scale decision architecture.

The framework integrates:

- geology,
- extraction technology,
- macroeconomics,
- finance,
- political economy,
- geopolitics,
- warfare economics,
- environmental systems,
- biology,
- psychology,
- psychiatry,
- statistics,
- machine learning,
- artificial intelligence.

The central insight is that economic activity is an emergent property of a larger coupled system linking physical resources, technological capability, institutions, geopolitical structures, environmental constraints, and information.

Planet Earth is therefore modeled as a stochastic, adaptive, learnable, and controllable dynamical system whose evolution may be analyzed through the unified mathematical framework developed in this treatise.

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Glossary

$\mathcal{M}$ Monetary–geopolitical manifold.

x Manifold state.

T_E Extraction technology stock.

T_M Military technology stock.

Γ Recoverability function.

Π Resource profit.

Y Aggregate output.

S_C State capacity.

G_P Geopolitical power.

M_S Military capability.

E Environmental damage.

B Biodiversity.

W Social welfare.

z Planetary state vector.

The End